

Frequency Distribution

Weight Data of 40 Adults (pounds) - Table, Histogram, Frequency Polygon, Ogive

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The Raw Data

Weight (in pounds) of 40 adults, collected with no order. Raw data must be organized numerically before it can be analyzed.

138	164	150	132	144	125	149	157
146	158	140	147	136	148	152	144
168	126	138	176	163	119	154	165
146	173	142	147	135	153	140	135
161	145	135	142	150	156	145	128

Step 1 - Range

Highest value = **176** • Lowest value = **119** • $n = 40$

Range = $176 - 119 = 57$ pounds

Step 2 - Number of Classes (Sturges)

$$k = 1 + 3.322 \log n$$

$k = 1 + 3.322 \times \log 40 = 1 + 3.322 \times 1.602 = 1 + 5.32 = 6.32 \rightarrow$ round up to **7 classes**

Step 3 - Class Width

$$c = (\text{highest} - \text{lowest}) / k$$

$c = 57 / 7 = 8.14 \rightarrow$ round up to **9**

Lowest class limit = lowest value = **119**. Width 9 guarantees all data fits (including 176).

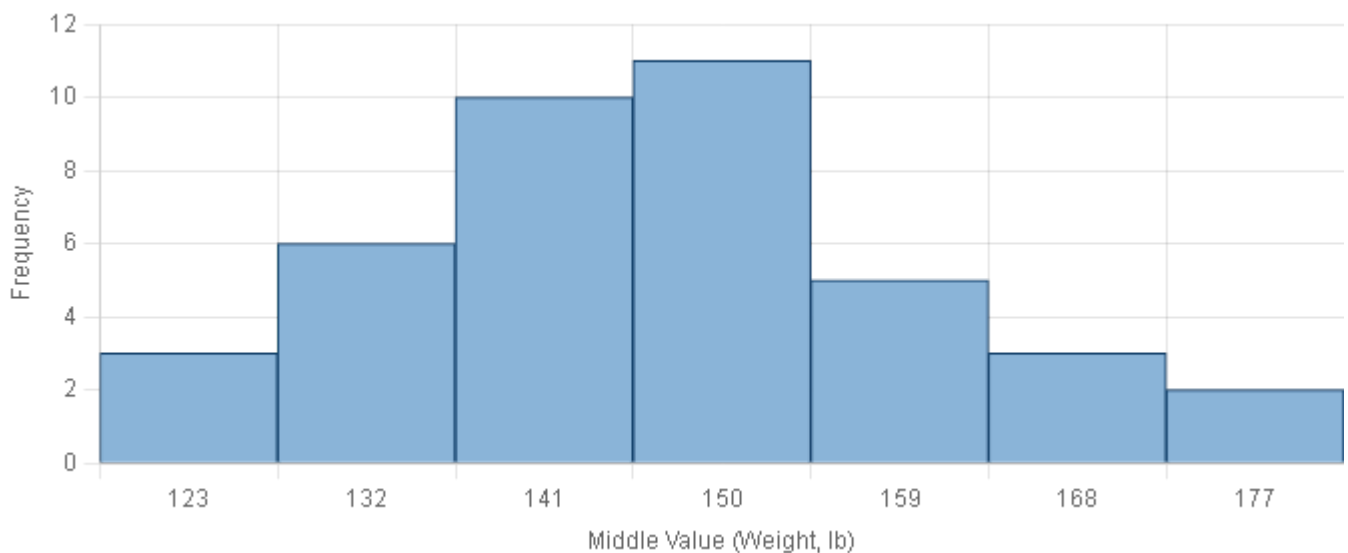
Step 4 - Frequency Distribution Table

Each value is tallied into its class. **Class limits** are real boundaries (limit ± 0.5) so no value falls into two classes. **Middle value** = (lower + upper)/2. **Cumulative frequency** = running total.

Class Interval	Class Limits	Middle Value	Frequency (f)	Cumulative f
119 – 127	118.5 – 127.5	123	3	3
128 – 136	127.5 – 136.5	132	6	9
137 – 145	136.5 – 145.5	141	10	19
146 – 154	145.5 – 154.5	150	11	30
155 – 163	154.5 – 163.5	159	5	35
164 – 172	163.5 – 172.5	168	3	38
173 – 181	172.5 – 181.5	177	2	40
Total			40	

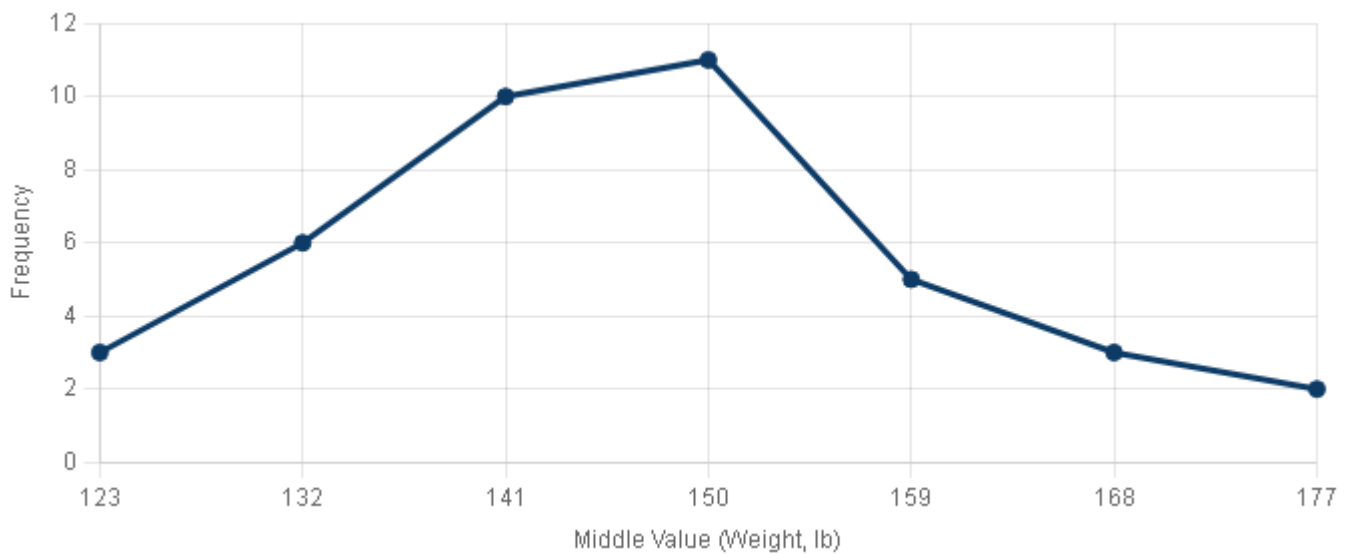
Histogram

Bars: x = middle value of each class, y = frequency. No gaps between bars (continuous data).

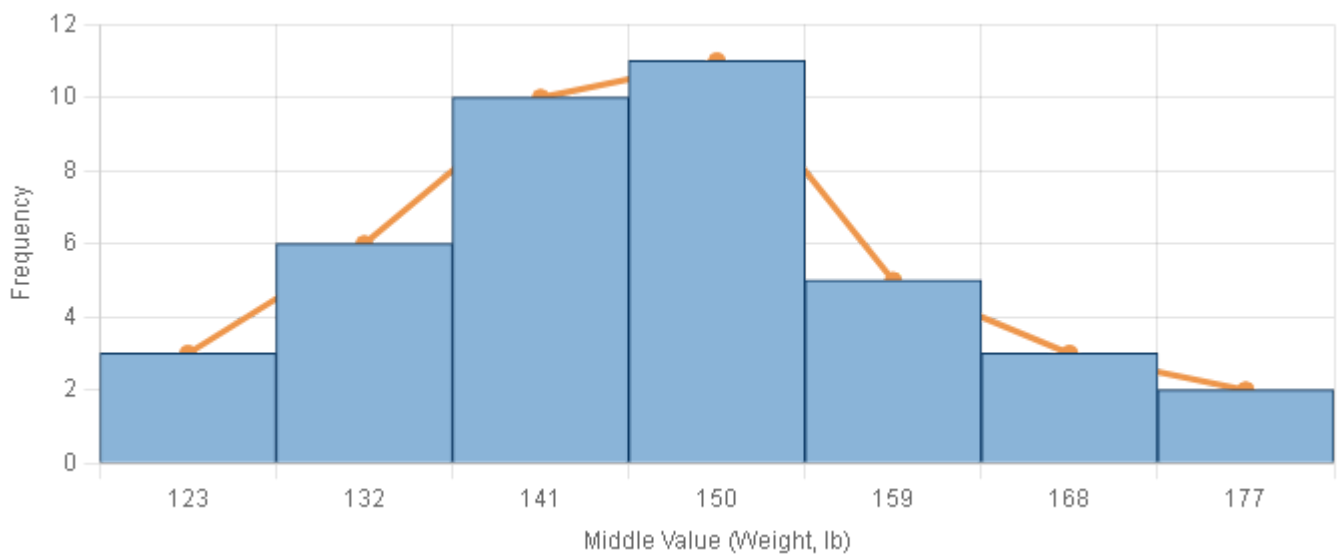


Frequency Polygon (Frequency Diagram)

Line graph connecting the middle value of each class against its frequency.

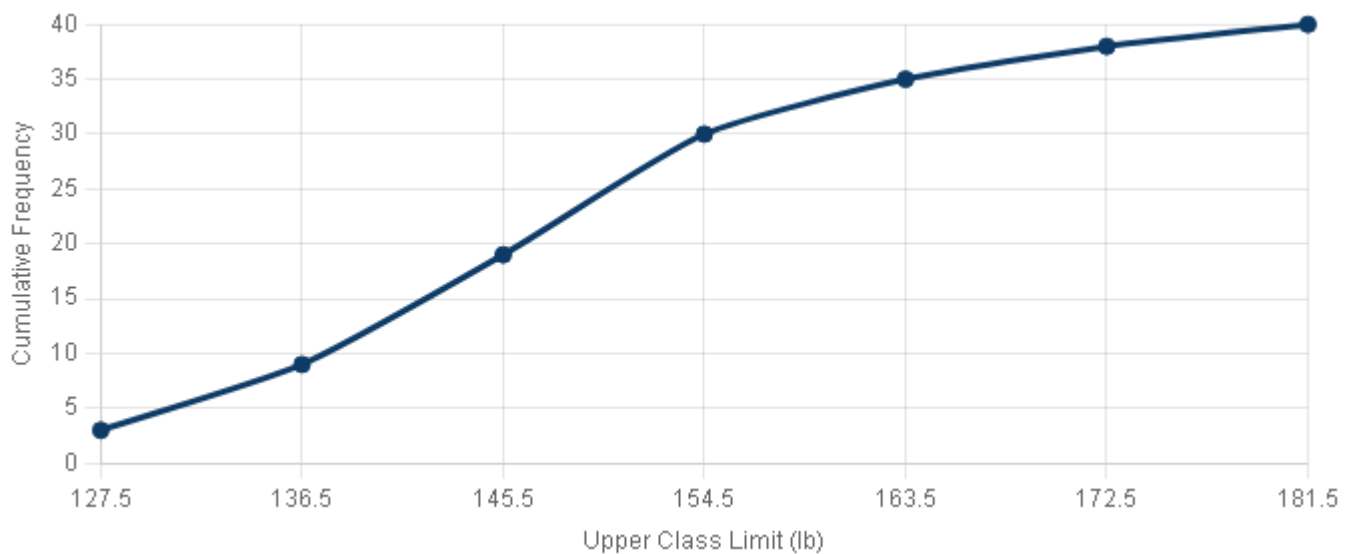


Histogram + Polygon Combined



Ogive (Cumulative Frequency Curve)

Plots cumulative frequency against the **upper class limit** of each class. Shows how many adults weigh below a given value.



Reading the Ogive

- 30 of 40 adults weigh under **154.5 lb**
- Median weight (20th value) falls in the **146–154** class
- Curve always rises, ending at $n = 40$

Summary

1. Find range → 2. Sturges for class count (7) → 3. Width (9) → 4. Tally into table → 5. Draw histogram, polygon, ogive.

The distribution peaks at the **146–154** class, most adults cluster around 150 lb.

Center & Dispersion - Setup

Using the frequency table from Chapter 2 ($n = 40$, class width $c = 9$). Grouped formulas use the class middle value and cumulative frequency.

Class Interval	Class Limits	Middle Value (x)	Frequency (f)	f · x	Cumulative f
119 – 127	118.5 – 127.5	123	3	369	3
128 – 136	127.5 – 136.5	132	6	792	9
137 – 145	136.5 – 145.5	141	10	1410	19
146 – 154	145.5 – 154.5	150	11	1650	30
155 – 163	154.5 – 163.5	159	5	795	35
164 – 172	163.5 – 172.5	168	3	504	38
173 – 181	172.5 – 181.5	177	2	354	40
Total			40	5874	

Mean

$$\bar{x} = \Sigma (f \cdot x) / \Sigma f$$

$$\bar{x} = 5874 / 40 = \mathbf{146.85}$$

Median

$$Md = L + ((n/2 - F) / f) \cdot c$$

20th value falls in class **146–154**. $L = 145.5$, $F = 19$, $f = 11$, $c = 9$.

$$Md = 145.5 + ((20 - 19) / 11) \cdot 9 = 145.5 + 0.82 = \mathbf{146.32}$$

Mode

$$Mo = L + (a / (a + b)) \cdot c$$

Highest frequency is class **146–154** ($f = 11$). $L = 145.5$, $a = 11 - 10 = 1$, $b = 11 - 5 = 6$, $c = 9$.

$$Mo = 145.5 + (1 / 7) \cdot 9 = 145.5 + 1.29 = \mathbf{146.79}$$

Quartile (Q1)

Position of Q_1 is the $(n/4)$ -th value.

$$Q_1 = L + ((n/4 - F) / f) \cdot c$$

Q1 → pos = $40 / 4 = 10$ → class **137–145** ($L = 136.5$, $F = 9$, $f = 10$, $c = 9$)

$$Q1 = 136.5 + ((10 - 9) / 10) \cdot 9 = 136.5 + 0.9 = \mathbf{137.40}$$

Lower quarter of adults weigh below 137.4 lb.

Decile & Percentile

Position of D_x is $(n \cdot x/10)$, of P_x is $(n \cdot x/100)$. (The brief lists "15th decile" but only D1–D9 exist; D5 is shown, which equals the median.)

$$D_x = L + ((n \cdot x/10 - F) / f) \cdot c$$

D5 → pos = $40 \times 5/10 = 20$ → class **146–154**

$$D5 = 145.5 + ((20 - 19) / 11) \cdot 9 = \mathbf{146.32}$$

$$P_x = L + ((n \cdot x/100 - F) / f) \cdot c$$

P10 → pos = $40 \times 10/100 = 4$ → class **128–136** ($L = 127.5$, $F = 3$, $f = 6$)

$$P10 = 127.5 + ((4 - 3) / 6) \cdot 9 = \mathbf{129.00}$$

Variance & Standard Deviation

Spread of the data around the mean (146.85). Sample variance divides by $n - 1$.

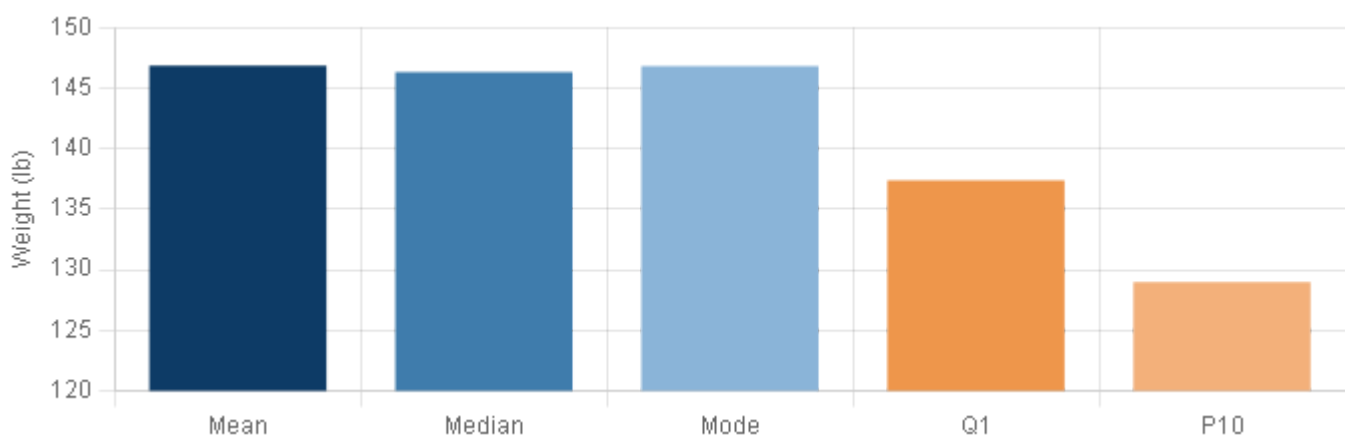
x	f	$x - \bar{x}$	$(x - \bar{x})^2$	$f(x - \bar{x})^2$
123	3	-23.85	568.82	1706.47
132	6	-14.85	220.52	1323.13
141	10	-5.85	34.22	342.22
150	11	3.15	9.92	109.15
159	5	12.15	147.62	738.11
168	3	21.15	447.32	1341.97
177	2	30.15	909.02	1818.05
$\Sigma f(x - \bar{x})^2$				7379.10

$$s^2 = \Sigma f(x - \bar{x})^2 / (n - 1)$$

$$s^2 = 7379.10 / 39 = \mathbf{189.21}$$

$$s = \sqrt{s^2} = \sqrt{189.21} = \mathbf{13.76}$$

Summary of Measures



Mean 146.85, Median 146.32, Mode 146.79 are nearly equal → distribution is close to symmetric, centered around **147 lb**. Standard deviation 13.76 lb shows typical spread from the mean.